FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no.20

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Recap *of previous lecture*

LAPLACE TRANSFORM



Topics to be covered

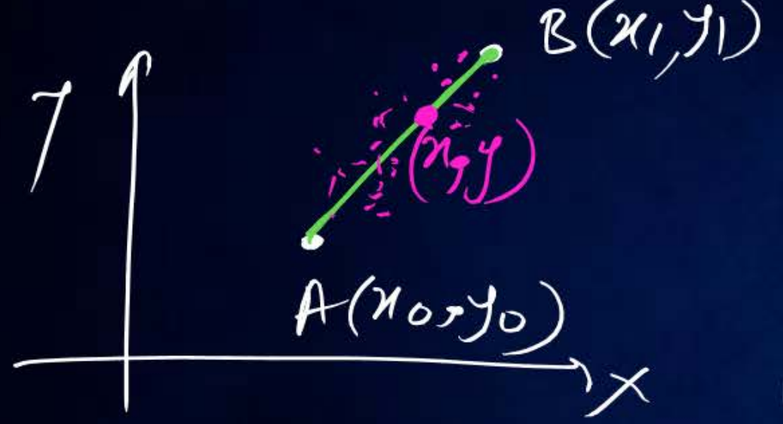
"INTERPOLATION"

(Mainly for CE/ME)



Linear Interpolation —

Equ^n of straight line passing through A & B



$$y - y_0 = \left(\frac{y_1 - y_0}{x_1 - x_0}\right)(x - x_0)$$

$$\boxed{y = y_0 + (x - x_0)\left[\frac{y_1 - y_0}{x_1 - x_0}\right]}$$

Let $x_0 < x < x_1$

Note — $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots\cdots$

Linear Approximation

Here $(x_0, y_0) = (a, f(a))$

$(x_1, y_1) = (x, f(x))$

$$\Rightarrow y = y_0 + (x - x_0)\left(\frac{dy}{dx}\right)_{x=a} = \boxed{y_0 + (x - x_0)\left(\frac{dy}{dx}\right)_{(x_0, y_0)}}$$

Q. Consider that we have given two points (3,4) & (6,8) then $y(5) = ?$

Sol: $(x_0, y_0) = (3, 4)$

$(x_1, y_1) = (6, 8)$

$\Rightarrow$

	x_0	x_1
$x =$	3	6
$y =$	4	8
	y_0	y_1

$$y = y_0 + (x - x_0)\left[\frac{y_1 - y_0}{x_1 - x_0}\right]$$

$$f(x) = 4 + (x-3)\left[\frac{8-4}{6-3}\right]$$

$$f(x) = 4 + \frac{4}{3}(x-3)$$

$$\therefore y(5) = f(5) = 4 + \frac{4}{3}(5-3) = 4 + \frac{8}{3} = \frac{20}{3} = 6.67$$

Q: Calculate the Estimated Height of Boy at 4th position;

Position x =	1	2	3	4	5
Height y =	3	4.5	5	?	6

we will Try to Interpolate b/w $x=3$ & 5

ie $x_0 = 3$ & $x_1 = 5$

$y_0 = 5$ & $y_1 = 6$

$$y = y_0 + (x - x_0)\left[\frac{y_1 - y_0}{x_1 - x_0}\right]$$

$$= 5 + (x-3)\left[\frac{6-5}{5-3}\right] = 5 + \frac{1}{2}(x-3)$$

so at $x = 4$,

$$y(4) = 5 + \frac{1}{2}(4-3) = 5.5$$

Q: The height of a plant changes according to the following chart;

Day	Height
1	0
3	4
5	8
7	12
9	16

Then Calculate estimated height of Plant on 4th day?

Sol: We will try to interpolate b/n 3rd day & 5th day?

i.e. $x_0 = 3$, $x_1 = 5$ & $y(4) = ?$

$y_0 = 4$, $y_1 = 8$

$$y = y_0 + (x - x_0)\left[\frac{y_1 - y_0}{x_1 - x_0}\right]$$

$$y = 4 + (x-3)\left[\frac{8-4}{5-3}\right] = 4 + (x-3)(2) = 2x - 2$$

i.e. $y = 2x - 2$

$y(4) = 6$ i.e. Height (4th day) $= 6$

Lagrange's Interpolation → Let given points are

x_0	x_1	x_2	x_3
y_0	y_1	y_2	y_3

then $y(x) = ?$

$$y = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}y_2$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}y_3$$

Quadratic Interpolation → When three points are given, we can use this concept,

$$y = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$$ for

x_0	x_1	x_2
y_0	y_1	y_2

Q. Consider the set of values

	x_0	x_1	x_2
$x=$	1	2	3
$y=$	3	8	15
	y_0	y_1	y_2

Then evaluate $y(2.5)$ using ① Q. Interpolation ② Linear Interpolation

$$y=\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}y_0+\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}y_1+\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}y_2$$

$$=\frac{(x-2)(x-3)}{(1-2)(1-3)}(3)+\frac{(x-1)(x-3)}{(2-1)(2-3)}(8)+\frac{(x-1)(x-2)}{(3-1)(3-2)}(15)$$

$$=\frac{3}{2}\left[x^2-5x+6\right]-8\left[x^2-4x+3\right]+\frac{15}{2}\left(x^2-3x+2\right)$$

$$\boxed{y=x^2+2x} \Rightarrow y(2.5)=\frac{25}{4}+5=\frac{45}{4}=11.25$$

② for L. Interpolation we will take $x_0=2$ & $x_1=3$

$y(2.5)=$ HW

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Thank
YOU